Discrete Mathematical Structures

CSET106

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Module 3

Recurrence Relations and Generating Functions

BENNETT UNIVERSITY
THE TIMES GROUP

Recurrence Relations

Recurrence Relation

A recurrence relation is an equation that recursively defines a sequence where each term is defined in terms of previous terms.

- Recurrence relations are often used to model sequences in mathematics and computer science.
- They can describe various phenomena such as growth rates, algorithms, and recursive processes.

Examples of Recurrence Relations

- **Example 1:** The Fibonacci sequence is defined by the recurrence relation:

$$F(n) = F(n-1) + F(n-2) \quad \text{for } n \geq 2$$

with initial conditions $F(0) = 0$ and $F(1) = 1$.

- **Example 2:** The factorial function $n!$ can be defined recursively as:

$$n! = n \cdot (n-1)! \quad \text{for } n \geq 1$$

with initial condition $0! = 1$.

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Generating Functions

Generating Function

A generating function is a formal power series that encodes information about a sequence.

- Generating functions provide a powerful tool for analyzing sequences, especially those defined by recurrence relations.
- They can be used to find closed-form expressions, calculate sums, and derive properties of sequences.

"A generating function is a device somewhat similar to a bag. Instead of carrying many little objects detachedly, which could be embarrassing, we put them all in a bag, and then we have only one object to carry, the bag." – George Pólya

Examples of Generating Functions of a Recurrence Relation

- **Example 1:** The generating function for the Fibonacci sequence is:

$$F(x) = \sum_{n=0}^{\infty} F(n)x^n = \frac{\text{[redacted]}}{1 - x - x^2}$$

This allows us to easily derive properties of the Fibonacci sequence.

- **Example 2:** The generating function for the factorial sequence is:

$$F(x) = \sum_{n=0}^{\infty} n!x^n = e^x$$

This is a result of the Maclaurin series for e^x.

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Recurrence Relations and Generating Functions

- **Example 1:** Consider the sequence defined by the recurrence relation:

$$a_n = 2a_{n-1} + 3a_{n-2} \quad \text{for } n \geq 2$$

  with initial conditions $a_0 = 1$ and $a_1 = 2$.

- The generating function for this sequence is: $A(x) = \sum_{n=0}^{\infty} a_n x^n$

  Multiplying the recurrence relation by x^n and summing over n, we get:

$$A(x) - a_0 - a_1 x = 2x(A(x) - a_0) + 3x^2 A(x)$$

  Solving for $A(x)$, we find: $A(x) = \dfrac{1}{1 - 2x - 3x^2}$.

[redacted]

- **Example 2:** Consider the sequence defined by the recurrence relation:

$$b_n = b_{n-1} + 2^n \quad \text{for } n \geq 1$$

with initial condition $b_0 = 0$.

- The generating function for this sequence is: $B(x) = \sum_{n=0}^{\infty} b_n x^n$.

Multiplying the recurrence relation by x^n and summing over n, we get:

$$B(x) - b_0 = x(B(x) - b_0) + \sum_{n=1}^{\infty} 2^n x^n$$

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- **Exercise Problem 1:** Consider the sequence defined by the recurrence relation:

$$c_n = 3c_{n-1} - 2c_{n-2} \quad \text{for } n \geq 2$$

with initial conditions $c_0 = 1$ and $c_1 = 3$.

- An[redacted].
- **Exercise Problem 2:** Consider the sequence defined by the recurrence relation:

$$d_n = d_{n-1} + d_{n-2} + 2^n \quad \text{for } n \geq 2$$

with initial conditions $d_0 = 0$ and $d_1 = 1$.

- Ans.[redacted]

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Thank You.